

2026 Y6 Term 3 Timed Practice Paper 1 – Solutions

Qn	Solutions
1a	EquipmentRental - modelNumber - quantity - rentalCostPerDay - numberOfDaysRented + get_modelNumber() + get_quantity() + get_rentalCostPerDay() + get_numberOfDaysRented() + set_modelNumber() + set_quantity() + set_rentalCostPerDay() + set_numberOfDaysRented() + calculateRentalCost() + displayDetails() [not required] PhotographyEquipment - storageDeviceIncluded - tripodIncluded + get_storageDeviceIncluded() + get_tripodIncluded() + set_storageDeviceIncluded() + set_tripodIncluded() + calculateRentalCost() [not required] + displayDetails() [not required] AudioEquipment - technicianName - venue + get_technicianName () + get_venue() + set_technicianName() + set_venue() + calculateRentalCost() [not required] + displayDetails() [not required] <pre> classDiagram class EquipmentRental { -modelNumber -quantity -rentalCostPerDay -numberOfDaysRented +get_modelNumber() +get_quantity() +get_rentalCostPerDay() +get_numberOfDaysRented() +set_modelNumber() +set_quantity() +set_rentalCostPerDay() +set_numberOfDaysRented() +calculateRentalCost() +displayDetails() } class PhotographyEquipment { -storageDeviceIncluded -tripodIncluded +get_storageDeviceIncluded() +get_tripodIncluded() +set_storageDeviceIncluded() +set_tripodIncluded() +calculateRentalCost() +displayDetails() } class AudioEquipment { -technicianName -venue +get_technicianName() +get_venue() +set_technicianName() +set_venue() +calculateRentalCost() +displayDetails() } EquipmentRental < -- PhotographyEquipment EquipmentRental < -- AudioEquipment </pre> displayDetails method not required as question did not ask for polymorphism.
1b	A superclass is used to store common properties and methods that can be shared by multiple subclasses. This avoids duplication and improves code reusability. Example from the scenario: EquipmentRental is the superclass. PhotographyEquipment and AudioEquipment inherit from it.

1c	Encapsulation is used to protect data by restricting direct access to object attributes and allowing access through methods. Example: The attribute rentalCostPerDay can be made private. A method such as setRentalCost() can be used to validate the value before updating it.
1d	Add a new property to the PhotographyEquipment class: photographerRequired Add method calculateRentalCost() in PhotographyEquipment so that the rental price calculation depends on whether a photographer is required. For example: additional charges may be added if photographerRequired = True
1e	Polymorphism allows the same method name to behave differently in different classes. Example: Both PhotographyEquipment and AudioEquipment have a method called calculateRentalCost(). However, each subclass can calculate the rental cost differently. For example: PhotographyEquipment may include photographer charges. AudioEquipment may include technician or venue-related charges. Thus, the same method name performs different implementations depending on the object type.
2a	Pass 1: [1, 5, 7, 8, 4, 2] Pass 2: [1, 2, 7, 8, 4, 5] Pass 3: [1, 2, 4, 8, 7, 5]
2b	$O(n^2)$ Justification • The algorithm uses a nested loop. • The outer loop runs approximately n times. • For each pass of the outer loop, the inner loop also runs up to n times. • Therefore, the total number of comparisons grows approximately as: $n \times n = n^2$ Hence, the algorithm has quadratic time complexity.
2c	[1000, 2000, 3000] This is erroneous because MinVal is initialized to 999, so no value in the list is smaller than 999.
2d	Instead of initializing: MinVal $\leftarrow$ 999 initialize MinVal using the first unsorted element: MinVal $\leftarrow$ MyList[i] This ensures the algorithm works correctly for any positive integers, including values larger than 999.

2e	For the sorted list: [1, 2, 4, 5, 7, 8] Linear Search • Checks each element one by one. • To find 8, up to 6 comparisons are needed. Binary Search • Checks the middle element first. • The search space is repeatedly halved. • 8 can be found in about 3 comparisons. Therefore, binary search is more efficient because it requires fewer comparisons.
2f	Linear Search • Maximum comparisons increase directly with list size. • For 600 elements, up to 600 comparisons may be needed. Binary Search • Comparisons increase much more slowly because the list is repeatedly halved. • For 600 elements, maximum comparisons increase only from 3 to 10. Thus, binary search scales much better for large lists.
2g	A recursive algorithm calls itself repeatedly to solve smaller versions of the same problem. A recursive algorithm must contain a base case to stop further recursive calls and prevent infinite recursion.
3a	The Purchases table is not in 3NF. Composite primary key consists of CustomerID, AlbumID, and Datetime. The customer details: • CustomerName • CustomerPhoneNo depend on CustomerID, rather than on the whole composite key. This results in data redundancy because the same customer information is repeated whenever the customer makes multiple purchases.

3b	 Alternative:
3c	Albums(<u>AlbumID</u>, AlbumName, Artiste, Genre, Price) Purchases(<u>CustomerID</u>, <u>AlbumID</u>, <u>Datetime</u>, Quantity) Customers(<u>CustomerID</u>, CustomerName, CustomerPhoneNo) Primary key: Underline Foreign key: Dashed underline Alt: Introduce PurchaseID as PK in Purchases table Alternative: Customers(<u>CustomerID</u>, CustomerName, CustomerPhoneNo) Albums(<u>AlbumID</u>, AlbumName, Artiste, Genre, Price) PurchaseItem(<u>PurchaseID</u>, <u>AlbumID</u>, Quantity) Purchases(<u>PurchaseID</u>, <u>CustomerID</u>, Datetime)
3d	<pre>SELECT Albums.AlbumName, Albums.Price FROM Albums INNER JOIN Purchases ON Albums.AlbumID = Purchases.AlbumID INNER JOIN Customers ON Purchases.CustomerID = Customers.CustomerID WHERE Customers.CustomerName = 'Jane Lee' AND Albums.Genre = 'K-pop';</pre> Alternative: <pre>SELECT Albums.AlbumName, Albums.Price FROM Albums, Purchases, Customers WHERE Albums.AlbumID = Purchases.AlbumID AND Purchases.CustomerID = Customers.CustomerID AND Customers.CustomerName = 'Jane Lee' AND Albums.Genre = 'K-pop';</pre>
3e	For Customer Reviews: NoSQL is required because reviews are unstructured (or semi-structured) data; it provides a flexible/dynamic schema that can store varying data types (e.g., text, ratings, or media) without a predefined table structure. For Financial Transactions: A relational database is required because transaction data must be immediately consistent; it enforces ACID properties to ensure that all nodes are updated simultaneously and account balances remain 100% accurate.

4a	Conditions	1	2	3	4	5	6	7	8	
	Is customer a member?	Y	Y	Y	Y	N	N	N	N	
	Buy Business Class?	Y	Y	N	N	Y	Y	N	N	
	Is it off-peak period?	Y	N	Y	N	Y	N	Y	N	
	Actions									
	20% discount	X								
	10% discount		X	X	X	X				
	No discount						X	X	X	
4b	Conditions	1	2	3	4	5	6			
	Is customer a member?	Y	Y	Y	N	N	N			
	Buy Business Class?	Y	Y	N	Y	Y	N			
	Is it off-peak period?	Y	N	-	Y	N	-			
	Actions									
	20% discount	X								
	10% discount		X	X	X					
	No discount					X	X			
4c	FUNCTION getDiscount(isMember, isBusiness, isOffPeak) IF isMember THEN discount ← 0.10 IF isBusiness AND isOffPeak THEN discount ← 0.20 ENDIF ELSE IF isBusiness AND isOffPeak THEN discount ← 0.10 ELSE discount ← 0.00 ENDIF ENDIF RETURN discount ENDFUNCTION									
4d	1. Disclosure of Personal Data – recipients’ email addresses The administrator revealed the email addresses of all recipients in the mass email. This exposed personal data to other recipients without consent.									
	2. Unauthorised Sharing of Personal Documents – Passport Verification Form The administrator sent another customer’s “Passport Verification Form” to an applicant. This disclosed confidential personal data to an unauthorised person.									
4e	1. Mass Email – The administrator should have used the BCC (Blind Carbon Copy) field so recipients could not see each other’s email addresses.									
	2. Passport Verification Form – The administrator should have used a blank template or removed all personal information before sending the document.									

5a	A protocol is a set of rules that govern how data is transmitted and received between devices on a network.
5b	It reduces complexity by allowing each layer to focus on a specific function independently of other layers.
5ci	HTTP is used to transfer web pages and related resources between a web browser and a web server.
5cii	The client (browser) sends an HTTP request to the web server for a resource (e.g., a webpage). The server processes the request and sends back an HTTP response containing the requested data or an error message.
5di	In client-server, there is a dedicated central server (centralised) providing resources. In P2P, there is no dedicated server (decentralised) and each node can act as both a client and a server.
5dii	A business needs a central point to manage security, backups, or updates to ensure data integrity and professional standards.
5e	When a URL is entered, the browser sends a request to a DNS server. The DNS server looks up and translates the domain name of the URL into an IP address. The IP address is returned to the browser, allowing it to locate and connect to the correct web server.